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EEG Signal processing for Detection of Colour Vision Deficiencies

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Abstract: The clinical assessment methods of detection of colour blindness are prone to subjective bias, human errors and lack of consistency. Also, heterogeneous nature of vision deficiencies makes manual detection difficult. Hence, objective methods based on physiological signals can be used. Various studies have been undertaken to analyse the effect that colour perceived has on EEG signals and as a result can be extended to detection of colour blindness. The results of this survey indicate that there are significant differences produced in EEG signals due to changes in colour as well as due to differences in colour perception. EEG signals also differ between healthy and colour vision deficient individuals, thereby providing a biomarker for detection. This biomarker can be exploited using machine learning algorithms for detection of colour related vision deficiencies.

Introduction

Colour blindness is one of the common vision disorders, particularly prevalent in men. It affects almost 8% of total male population of the world (1). Colour blindness occurs due to abnormalities in cone cells present in retina. There are three types of cones- red, green and blue. As different persons have abnormal renditions of different colours based on which type and number of cone cells are affected, colour vision deficiencies are heterogeneous in nature and can be either anomalous trichromacy, dichromacy or even monochromacy (2).

Human body activity is reflected by various physiological responses like EEG, GSR, EMG, ECG, skin temperature, retinal electrophysiological response and respiration rate etc. These responses can be recorded using different sensors and devices. These responses are affected by external stimuli as well as by emotions, physical activity and mental states due to changes in physiological activity. As such, various physiological responses show variations under the influence of abnormal conditions (3)-(5). Colour vision deficiencies also alter the physiological response of the body, specifically the EEG signals originating from the brain (6)-(8). The changes in these responses can provide biomarkers for colour vision deficiencies.

Different features of physiological signals must be extracted and analysed for the detection of intensity, prevalence or type of any aberration. Manual analysis of physiological signals is very difficult and requires specialized expertise and skill. Manual analysis is also very time-intensive and is prone to human error and subjective bias (9)-(11).



a). Normal Vision



b). Monochromacy



c). Tritanopia



d). Deuteranopia



e). Protanopia

Figure 1. Different types of Colour Vision Deficiencies (1)

Also, physiological signals have intra-individual differences which adds to the complexity (12), (13). The heterogeneous nature of vision also makes manual detection even more difficult. As such, machine learning has been extensively used for the analysis of physiological signals in different applications. Machine learning methods eliminate subjective bias and human errors and are relatively very consistent. So, machine learning can be applied on physiological data for various application areas related to colour vision deficiencies (14), (15).

The purpose of this survey is to highlight recent work towards colour vision deficiency detection using EEG signals and their potential use in developing brain-computer interfaces. The rest of the paper contains Literature Survey in Section 2, Section 3 introduces the basic framework of colour vision deficiency using EEG signals, Section 4 highlights the various challenges encountered in detection of colour vision deficiencies using EEG signals and Section 5 represents conclusion drawn from the study.

Literature Survey

Bieber et al. used EEG signals of male infants to measure Evoked Potentials for detection of colour vision deficiency in 4-week and 8-week old infants. The authors used EEG signals with a vector voltmeter to analyse the effects of cone-isolating stimuli on EEG signals. The results showed remarked differences between the response of colour deficient and healthy infants (16). [16].

Machado et al. proposed a model that tried to handle the heterogeneous nature of colour vision deficiencies by simulating anomalous trichromacy as well as dichromacy. The model used the changes in the absorption of spectrum by photopigments present inside cone cells. It was implemented in MATLAB (17).

Rasheed et al. used RGB stimuli to observe changes in frequency spectrum of EEG signals due to perceived colour. The authors used mean event-related potential to distinguish between colours. EEGLAB and MATLAB were used for processing and analysis of EEG signals acquired during a colour-viewing task. There were differences in frequency spectrum of colours presented to viewer (18).

Thomas and Umamaheshwari used RGB along with 12 different shades as a stimuli during a viewing task and EEG signals were recorded. The study indicates that there is a difference between energy of EEG signals, depending whether right or wrong colour is selected. For right answers, the energy tends to increase while for wrong answers, there is decrease in energy of EEG signal. Principal Component Analysis was used process the signals in MATLAB (19).

Arnab and Rimita carried out a study to detect colour perceived through EEG signals. The authors used red, yellow, blue and green colour stimuli in a viewing task and subsequently EEG signals were recorded. Each colour perceived rendered different power spectrum density. This provides a biomarker for detection of colour vision deficiencies using EEG signals. An Interval-Type II fuzzy classifier was used for automatic detection (20).

Thomas et al carried out a project to develop a device which could analyse the effects of colour vision deficiencies on EEG signals, thereby helping in detection of colour blindness through EEG signals. The authors used features like entropy and energy of EEG signals captured during a colour viewing task to discriminate between colour blind and healthy subjects. Beagle board and Python were used to develop the device (7).

Tcheslavski et al proposed a method based on multivariate temporal response functions to differentiate colours perceived by an individual. EEG signals were collected during a free viewing task. The results of the study show that maxima and minima of mTRFs obtained through Fp1 electrode of EEG differ with colour perceived by the individual during shorter period of visual stimulation (21).

Ana and Anabela designed a test based on Ishihara colour blindness test to analyse the effect of colour on EEG signals, wherein shapes of objects are presented in colour blocks to participants for identification and their EEG is recorded. A Mindwave EEG sensor was used for processing the EEG signals into digital form and makes use of eSense Algorithm for processing. The results of the study show that EEG signals vary with changes in contrast of colours and also with changes in colour combinations (8).

Wicaksono et al. used event-related potential from EEG signals collected during Ishihara visual stimuli to distinguish between normal vision and partial colour-blind individuals. The study revealed that there are differences in GFP, Brain Topography Analysis and ERP between normal and partial colour blind subjects at various values of latency (15).

Xiaowei et al. proposed a method to diagnose colour-vision deficiencies from EEG signals based on steady-state visual evoked potentials. Canonical Correlation Analysis was used for EEG processing and Colour Vision Severity Index based on activation of cone cells was proposed to distinguish between normal and deficient colour vision. The results show that Colour Vision Severity Index was different in people having different types of colour vision deficiencies. This method also handled anomalous trichromacy and dichromacy, particularly protanopia and deuteranopia (22).

Roy et al. analysed the EEG signal behaviour with changing colour stimuli. They used non-linear approaches of time-series data processing such as MFDFA and MFDXA. The study shows that the EEG complexity is different for different colours perceived. Also, lobe correlation is lower for VIBGYOP colours as compared to grey shade (6).

Duru and Alobaidi also studied changes in EEG activity with change in colour perceived by an individual. The authors used shapes in red, green and blue colours to evoke EEG response. Band power of delta, theta, alpha, beta and gamma bands of EEG signals was used along with features obtained from inverse solution of EEG waves. The results reveal that a significant effect of colour was observed in different electrodes. Machine learning algorithms like SVM, KNN, and Quadratic Discriminant were deployed for automatic classification (23).

Jasper et al used scalp EEG to decode visual colour and study changes in EEG activity due to colour stimuli. PCA in MATLAB was used to process the EEG signals collected during a viewing session wherein different colours were presented in different orientations. The study revealed that it is possible to decode colour perceived using EEG signals. EEG signals also varied due to luminance but were mostly altered by shifts in colour hues. LDA algorithm was used for classification (24).

Ekhlasi et al used frequency features like mean absolute value of delta, theta, alpha and beta frequency bands of 1 second EEG signals for detection of colour vision deficiency. The signals were collected during 38-plate Ishihara's Test. The study showed that average power frequency of different power bands of EEG signals varied with change in stimulus. KNN classifier was used for detection of colour blind subjects (14).

TABLE I. Summary of Literature Survey

Ref	Year	Purpose	No. of Participants	Technique/Algorithm /Method Used	Key Findings
(17)	2009	To Handle the heterogeneous nature of colour vision deficiencies	17 normal males, 13 CVD males.	MATLAB	Absorption spectrum changed with CVD type.
(18)	2010	To study changes in frequency spectrum of EEG signals due to perceived colour	7 normal subjects.	EEGLAB and MATLAB	Frequency spectrum of EEG signals changed with change in perceived colour.

(19)	2016	Effect on EEG signals due to colour choice	3 subjects	Principal Component Analysis, MATLAB	difference between energy of EEG signals, depending whether right or wrong colour is selected.
(20)	2016	To detect colour perceived using EEG signal analysis.	Not Mentioned	Interval-Type II fuzzy classifier	Power spectrum density varied between different colour stimuli
(7)	2017	To develop a device which could detect CVD through EEG	Not Mentioned	Beagle Board, Python	--
(21)	2018	To differentiate colours perceived using EEG	9 male and 10 female subjects	mTRF Analysis	maxima and minima of mTRFs differ with colour perceived by the individual.
(8)	2020	To study effect of colour on EEG signals.	8 males and 20 females	Mindwave EEG Sensor, eSense Algorithm	EEG signals vary with changes in contrast of colours and also with changes in colour combinations
(15)	2020	To distinguish between normal vision and partial colour-blind individuals using EEG.	21 normal males and 4 partial CVD males	GPF, Brain Topography Analysis, ERP Analysis	Differences in values between normal and partial colour blind subjects at various values of latency
(22)	2020	To diagnose different types of colour-vision deficiencies from EEG signals.	8 anomalous trichromats and 7 dichromats	Canonical Correlation Analysis, Colour Vision Severity Index	Colour Vision Severity Index was different in people having different types of colour vision deficiencies
(6)	2021	To study change in behaviour of EEG due to change in colour.	16 subjects	MF DFA and MF DXA	EEG complexity is different for different colours perceived.
(23)	2021	To study changes in EEG activity with change in colour perceived by an individual	10 subjects	SVM, KNN, and Quadratic Discriminant Analysis	significant effect of colour was observed in different electrodes.
(24)	2021	To decode visual colour from EEG signals.	30 normal subjects	MATLAB, LDA Classifier	EEG signals varied due to change in colours.
(14)	2021	To detect CVD using EEG signals	10 normal vision and 10 CVD subjects	KNN Classifier	Mean values of frequency bands were different in normal and CVD subjects.

Methodology for Colour Vision Deficiency Detection using EEG Signals

As discussed in Section 2, changes in colour perception brings about a change in different features of EEG signals like mean frequency band values, complexity and other statistical features. The changes in EEG signals due to colour vision deficiencies provides an important biomarker for detection of these disorders. The objective nature of this detection scheme makes it free from subjective bias prevalent in clinical diagnosis. Colour vision deficiencies can be detected by classifying EEG signals using classification algorithms. The various steps that can be followed for achieving this are as below:

1. Signal Acquisition: Subjects are selected and EEG signals are recorded under different circumstances and environments.
2. Signal Pre-processing: EEG signals collected contain noise like muscle signals, eye movements etc. These need to be isolated.
3. Feature Extraction: Features of interest need to be extracted from the EEG signals. Various linear and non-linear features pertaining to time and frequency domains can be extracted.
4. Feature Selection: Appropriate features need to be selected to reduce feature-dimensionality. This helps in achieving more efficient results.
5. Classification: Models can be trained on the selected features which will group the EEG features into healthy and colour blind classes.

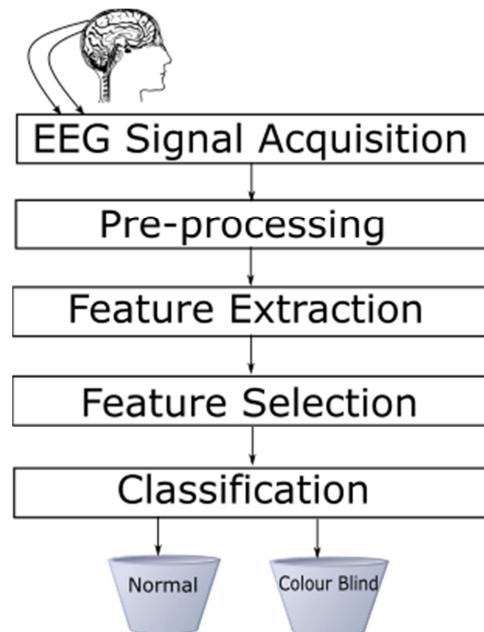


Figure 2. Methodology of Detection of Colour Vision Deficiency using EEG Signals

Challenges in Detection of Colour Vision Deficiency using EEG Signals

The research in detection of colour vision deficiencies using EEG signals has not seen much development despite the potential of early detection and elimination of undesirable factors inherent in subjective decoding. There are various challenges that arise in harnessing the discriminatory power of EEG signals for automatic detection of colour vision deficiencies. The most significant challenges are:

1. **Data Challenge:** The functionality and efficiency of machine learning models rely on data. In order to construct an efficient model, lots of data is needed for training. However, there is no dataset publicly available. Also construction of a new dataset will require resources and willing subjects. Also, as seen in Table 1, the highest number of CVD subjects included in a study is just 13. This hinders the development and research in developing brain-computer systems capable of producing reliable results.
2. **Emotions Challenge:** EEG signals are heavily altered by emotions. In detection of colour vision deficiencies using EEG, the emotional state of the subjects while data collection needs to be taken in consideration. Data collection sessions need to be properly designed and emotional state of participants be gauged.
3. **Demographics Challenge:** EEG signals are also dependent on demographic factors like age, gender, race etc. Incorporating all these factors in analysis will require lot of time, which should not be the case with computer-vision techniques.
4. **Ethical Challenge:** Collection of EEG data from patients and its subsequent usage presents a large ethical challenge. Various ethical considerations will be needed to generate a usable corpus of data. It includes permission from required authorities and willingness of test subjects.

Conclusion

Colour vision deficiency is a common vision deficiency prevalent particularly in men. Past research has showed that EEG can be used with machine learning algorithms for detection of human states and functions. Since colour vision deficiencies are also related to human brain like all human activities and states, EEG signals can be used for early and automatic detection of colour vision deficiencies. Machine learning algorithms can be used for automatic classification of EEG signals. However, there are certain challenges that need to be overcome. Still, detection of colour vision deficiencies with help of EEG and machine learning provides an upcoming avenue of research.

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